

Company: Barclays

Job Profile: SDE

Offer Type: Placement Only

CTC: 13.5 LPA

Location: Pune

Process Date: 26th July, 2024

Placement Process Overview

The placement process at Barclays involved three rounds, starting with an online assessment and progressing to two interview stages. On July 26th there was an online assessment, followed by interviews on July 29th. Out of 133 participants who appeared for the online assessment, 37 were shortlisted for the first technical interview. Subsequently, 21 individuals advanced to the final Tech + HR round,

Round 1: Online Assessment (90 minutes)

The Barclays online assessment was a three-part test consisting of MCQs, a SQL query, and a coding problem. The MCQs were the key differentiator for shortlisting.

1. Multiple Choice Questions (MCQs)

These questions tested foundational knowledge across various programming languages and concepts.

- **Core Language Concepts:** Questions involved predicting the output of short code snippets in languages like **C++** and **Java**. Topics included pointers, bitwise operators, control flow (**continue**), and variable scope.
- **Object-Oriented Programming (OOP):** This section focused on OOP principles, such as the difference between **abstract classes vs. interfaces** and **method overriding vs. overloading**. It also covered topics like the impossibility of overriding **private** and **static** methods and the order of class initialization.

- **Exception Handling (Java):** Questions evaluated an understanding of **try-catch-finally** blocks. A key point was knowing that a **return** statement in a **finally** block overrides any **return** from **try** or **catch**, and that **System.exit()** prevents the **finally** block from executing.

2. SQL Questions

This section included both a query and theoretical MCQs.

- **Queries:** The questions required you to apply your SQL knowledge to real-world scenarios. This involved using **multi-table joins** to combine data, filtering data by date using functions like **EXTRACT(MONTH FROM date)**, and translating complex business rules into a single **SELECT** query.
- **SQL Theory:** MCQs tested your knowledge of commands like **DROP vs. TRUNCATE vs. DELETE**. A crucial distinction was the difference between an **equi join** (which uses only **=**) and an **inner join** (a broader category of joins).

3. Coding Problem

The coding problem was a single, complex challenge that required careful logical thinking.

- **"String Generation":** The task was to create the lexicographically smallest string **S** based on a pattern string (**S1**) and a rule string (**S2**) of **'T'** (true) and **'F'** (false) constraints.
- **Solution Logic:** The recommended approach was to first satisfy all **'T'** constraints. Then, fill the remaining positions with the smallest possible character, **'A'**. Finally, iterate through all **'F'** constraints and minimally modify the string (e.g., change the rightmost **'A'** to a **'B'**) to break any invalid matches while preserving the lexicographical order. This problem went beyond a simple build-and-check method, demanding a more nuanced strategy to correct violations.

Round 2: Technical Round 1

Following the online assessment, 37 participants were shortlisted for the interview rounds. There were five interview panels, and I happened to be assigned to one of the more challenging ones. My interviewer was highly experienced and went into very detailed questions on every topic. This round lasted for approximately an hour.

The interview began with a standard introduction, after which I was asked to brief him about my projects. He then inquired about the programming languages I was comfortable with. Following this, he asked me to explain the concept of method overloading and to write a Java program demonstrating a real-world example, encouraging me to think beyond common examples like 'Area' or 'Vehicle'. He then moved on to an in-depth discussion about my projects. Noticing my significant frontend projects, he transitioned into detailed questions about React and HTML. He gave a scenario about creating a dropdown within another dropdown using React and asked about the distinction between a textbox and a textarea. Further questions included how to limit the number of words in an input field and how to display the count of letters already entered. While I couldn't recall every specific detail, I explained my approach clearly.

He then asked detailed HTML questions, presenting a scenario where he needed to input seven addresses in a form. He asked how I would extract the second address, impose a limit on the number of addresses, and manage IDs if they were assigned sequentially (e.g., 1 for the first address, 2 for the second). Specifically, he asked what would happen to ID 1 if the first address was deleted—would it be reassigned, or would a new one be created? He also wanted to know how I would track the total number of addresses.

After assessing my Java knowledge, he shifted to SQL. He asked if I knew about Procedures and Triggers, to which I confirmed. He then asked for an explanation and relevant queries. At this point, I honestly confessed that I hadn't practiced writing queries for procedures and triggers, but I explained the underlying concepts. I remained calm and continued to answer all his questions. He then inquired about the partitioning concept in DBMS and what partitioned tables are, asking me to write a query for it. Finally, he asked about different types of indexes in DBMS. The interview concluded with a question about the risks a bank might face and potential solutions.

I left this round feeling devastated and demotivated, convinced it had gone terribly wrong and almost on the verge of tears.

Round 3: Tech + HR Round

To my surprise, I was called for this final round, which shortlisted 21 candidates. This round lasted for about half an hour. The interviewer started by sharing the positive feedback he had received from my first interviewer, which was a huge relief and a pleasant surprise given how I felt after the first technical round.

He then asked me to describe my internship experience and elaborate on my projects, cross-questioning me further to see my understanding and contributions. Following this, he moved on to the HR-specific questions. He asked classic questions like "Why do you want to join Barclays?" and a challenging scenario: "If another company offered you double the salary, would you switch?" He also presented situations involving workplace ethics and problem-solving, such as: "If your manager gives you a task with an unrealistic deadline, how would you approach it?" and "You've completed your work for a strict weekend deadline and left for vacation, but midway, you realize you forgot to push an important update to GitHub. How would you resolve this?" Another thought-provoking question was: "If you were stuck on a project you built from scratch, and a friend helped you resolve a small error that had a big impact, and then you were nominated for an award for that project, would you accept the award or tell your team members about your friend's contribution? Discuss the pros and cons of each decision." He asked many more similar situational questions.

I ensured to be confident and true to myself in my responses, which seemed to satisfy the interviewer. At the end, I took the opportunity to ask him about his role and what Barclays expects from interns, specifically if prior knowledge of finance is required. With these questions, the interview concluded.

Areas for Improvement, Preparation, and General Tips

My selection was largely due to my ability to stay calm under pressure, even when faced with questions I couldn't fully answer. My honesty about my limitations, coupled with explaining my approach or conceptual understanding, showed integrity. My detailed preparation in OOPS, Java, HTML, React, SQL, and DBMS fundamentals was crucial, especially since MCQs were a deciding factor in the initial round. In the HR round, being confident, truthful, and demonstrating problem-solving skills with practical reasoning helped immensely. Engaging the interviewer and asking relevant questions also showed genuine interest and initiative.

Areas for Improvement: My biggest learning from this experience was the need for thorough practical application of concepts, especially in areas like SQL procedures and triggers, beyond just theoretical understanding. It's crucial to not only know *what* a concept is but also *how* to implement it with code or queries.

Preparation Strategy:

- **Online Assessment:** I primarily practiced MCQs from GeeksforGeeks to strengthen my foundational knowledge in OOPS, Java, C++, Objective-C, SQL, and DBMS. For coding, I focused on problems from LeetCode.
- **Technical Interview:** I focused on preparing OOPS, Java, SQL, and DBMS concepts in depth.
 - **Data Structures and Algorithms (DSA):** I referred to [Striver's SDE Sheet](#) and the [Blind 75 LeetCode Java - All Questions Solved](#) for coding practice.
 - **SQL:** [Top SQL 50 on LeetCode](#) was very helpful, along with [Lecture 10: Common SQL Queries for Interviews](#) and [SQL Bolt](#) for hands-on practice.
 - **DBMS:** [Complete DBMS in 1 Video \(With Notes\) || For Placement Interviews](#)
 - **OOPS & Java:** [Java OOPs in One Shot | Object Oriented Programming | Java Language | Place...](#) was a key resource. I also practiced Java MCQs from [GeeksforGeeks](#).
 - **Frontend (React/HTML):** For React, [Top 100 React JS Interview Questions and Answers](#) was a valuable resource, and I practiced detailed HTML concepts.
 - Practice SQL, DBMS, Java, OOPS, and React questions from InterviewBit.
- Ensure your resume clearly highlights your projects, particularly showcasing your role and contributions, and the technologies used. Be prepared to discuss every detail mentioned on your resume, as interviewers will cross-question. Don't include anything you're not confident about.

Important Points to Remember on the Day of the Interview:

- Regardless of how challenging the questions get, maintaining your composure is key. A small, calm smile can convey composure and confidence. Don't give up, even if a round feels like it went poorly.
- Always have a pen and paper handy for jotting down thoughts or working through problems.
- Keep your phone screen facing downwards to avoid any distractions.
- Sit straight, maintaining good posture, as it reflects engagement and confidence.

- Strive to make the interview a two-way conversation, engaging with the interviewer. Think out loud while solving problems; it helps the interviewer understand your thought process.
- If you don't know an answer completely, it's better to honestly confess and explain what you do know or how you would approach finding a solution, rather than bluffing.
- At the end, always ask thoughtful questions about the role, team, company culture, or expectations. This shows genuine interest.

By keeping these points in mind, and remembering to stay confident and true to yourself, you can significantly improve your chances during the placement process.

ALL THE BEST!!

You can contact me on LinkedIn anytime you want!!